

CIRCUIT THEORY II - R4053 - 2025	
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T.P. No. 6: Synthesis of quadripoles	
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Exercise #1

Synthesize an RC quadripole in vacuum that meets:

$$\begin{aligned} -Y_{21} &= \frac{s(s+1)}{s+2} \\ Y_{22} &= \frac{s(s+2.25)}{s+2} \\ \frac{V_2}{V_1} &= \frac{(s+1)}{(s+2.25)} \text{ for } I_2=0 \end{aligned}$$

1) Synthesize an quadripole RC en vacío que cumpla con:

$-Y_{21} = \frac{s(s+1)}{s+2}$ y $Y_{22} = \frac{s(s+2.25)}{s+2}$ tal que $\frac{V_2}{V_1} = \frac{s+1}{s+2.25}$ con $I_2=0$

2) Obtén la estructura de polos y ceros de los parámetros

→ De $-Y_{21} = \frac{s(s+1)}{s+2}$, tiene "ceros" en $s=0$ y $s=-1$ y polo en $s=-2$

→ De $Y_{22} = \frac{s(s+2.25)}{s+2}$, tiene "ceros" en $s=0$ y $s=-2.25$ y polo en $s=-2$

Dado que los polos y "ceros" sobre eje s y el origen $\Rightarrow$ sistema disipativo

→ $\frac{-Y_{21}}{Y_{22}} = \frac{\frac{s(s+1)}{s+2}}{\frac{s(s+2.25)}{s+2}} = \frac{s+1}{s+2.25} = \frac{V_2}{V_1} \Big|_{I_2=0}$ gives

3) Aplica método gráfico, debes comenzar con $\frac{V_2}{V_1} = 1$ puesto que los ceros de Y_{22} producen polos en transpuesta $\frac{V_2}{V_1} \Big|_{I_2=0}$. Se pretende, además, eliminar los ceros de transpuesta $\frac{V_2}{V_1} \Big|_{I_2=0}$, utilizando resistencia parcial o completa según tipo de ceros finitos e infinitos.

The handwritten notes show the synthesis of an RC quadripole. It includes pole-zero plots for $-Y_{21}$ and Y_{22} , a partial fraction decomposition of the transfer function, and a corresponding circuit diagram with resistors and capacitors.

Pole-zero plots for $-Y_{21}$ and Y_{22} are shown, with poles at $s=-2$ and zeros at $s=0$ and $s=-1$ for $-Y_{21}$, and $s=0$ and $s=-2.25$ for Y_{22} .

The partial fraction decomposition is given as:

$$\frac{s+1}{s+2.25} = \frac{A}{s+2} + \frac{B}{s+2.25}$$

The circuit diagram shows a network of resistors and capacitors that realizes the transfer function.

3

Exercise #2

Synthesize a quadripole that complies with:

$$Z_{21} = \frac{s}{s^2 + 2}$$

$$Z_{22} = \frac{1 + 2s^2}{s(s^2 + 2)}$$

2) Sintetice un cuatropolo que cumpla con:

$$Z_{21} = \frac{s}{s^2 + 2} \quad y \quad Z_{22} = \frac{1 + 2s^2}{s(s^2 + 2)}$$

1º) Observe estructuras de Z_{21} y Z_{22}

→ De $Z_{21} = \frac{s}{s^2 + 2}$, tiene "Zero" en $s=0$ y polo en $s = \pm j\sqrt{2}$

→ De $Z_{22} = \frac{1 + 2s^2}{s(s^2 + 2)}$, tiene "Zero" en $s = \pm j\sqrt{2}$ y polo en $s=0$ y $s = \pm j\sqrt{2}$

Dado que hay polos y "zeros" sobre eje SW y el origen $\Rightarrow$ sintetizar en cascadas

2º) Aplique método gráfico, puesto comenzamos con Z_{21} o Z_{22} por interconectar:

circuito resultante:

3º) Puntos de partida para todo polo restante, así obtengo valores de componentes

$$Z_2 = Z_{22} - \frac{K_{01}}{s}, \quad K_{01} = \lim_{s \rightarrow 0} Z_{22}(s-0) = \lim_{s \rightarrow 0} \frac{1 + 2s^2}{s(s^2 + 2)} \cdot s = \frac{1 + 2 \cdot 0^2}{(0^2 + 2)} = \frac{1}{2} \cdot K_{01}$$

$$Z_2 = \frac{1 + 2s^2}{s(s^2 + 2)} - \frac{1/2}{s} = \frac{1 + 2s^2 - (s^2 + 2) \cdot (1/2)}{s(s^2 + 2)} = \frac{1 + 2s^2 - (s^2/2 + 1)}{s(s^2 + 2)} = \frac{(3/2)s^2}{s(s^2 + 2)} = \frac{(3/2)s}{s^2 + 2}$$

$$Z_2 = \frac{(3/2)s}{s^2 + 2}, \quad Y_2 = \frac{s^2 + 2}{(3/2)s}$$

húsaes

$$Y_4 = Y_2 - \frac{K_{02}}{s}, \quad K_{02} = \lim_{s \rightarrow 0} Y_2(s-0) = \lim_{s \rightarrow 0} \frac{s^2 + 2}{(3/2)s} \cdot s = \lim_{s \rightarrow 0} \frac{(s^2 + 2)}{(3/2)} = \frac{2}{3} \cdot (0^2 + 2) = \frac{4}{3}$$

$$Y_4 = \frac{s^2 + 2}{(3/2)s} - \frac{4/3}{s} = \frac{s^2 + 2 - (4/3)s^2}{(3/2)s} = \frac{(2/3)s^2 + 2}{(3/2)s} = \frac{(2/3)s^2 + 2}{(3/2)s}$$

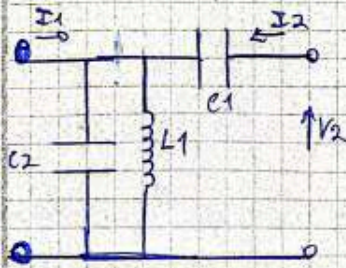
$$Y_4 = \frac{\frac{2}{3}(s^2+2) - 4/3}{s} = \frac{\frac{2}{3}s^2 + \frac{4}{3} - \frac{4}{3}}{s} = \frac{2}{3}s$$

$$Y_5 = Y_4 - K_{00} \cdot s, \quad K_{00} = \lim_{s \rightarrow \infty} Y_4 \cdot \frac{1}{s} = \lim_{s \rightarrow \infty} \frac{2}{3}s \cdot \frac{1}{s} = \frac{2}{3} = K_{00}$$

$$Y_5 = \frac{2}{3}s - \frac{2}{3}s = 0$$

10) Plantas circuitos con volúmenes de componentes

circuitos con componentes



$$C_1 = \frac{1}{K_{01}} = \frac{1}{1/2} = 2$$

$$C_2 = K_{00} = \frac{2}{3}$$

$$L_1 = \frac{1}{K_{02}} = \frac{1}{4/3} = \frac{3}{4}$$

5.4) Realiza verificación del que obtengo el parámetro Z_{22} trabajando por otro método planteando matrices o usando la definición de parámetro

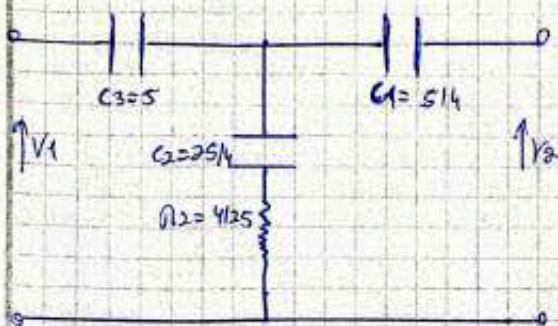
$$Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0} = \frac{1}{Y_{L1}C_2} + Z_{C1} = \frac{1}{\frac{1}{K_{01}C_2}} + \frac{K_{01}}{s}$$

$$Z_{22} = \frac{1}{\frac{1}{\frac{3}{4}s} + \frac{2}{3}s} + \frac{1/2}{s} = \frac{1}{\frac{4}{3s} + \frac{2}{3}s} + \frac{1}{2s}$$

$$Z_{22} = \frac{1}{\frac{4 + 2s^2}{3s}} + \frac{1}{2s} = \frac{3s}{4 + 2s^2} + \frac{1}{2s} = \frac{3s \cdot 2s + 2s^2 + 4}{(2s^2 + 4) \cdot 2s}$$

$$Z_{22} = \frac{8s^2 + 4}{(2s^2 + 4)2s} \cdot \frac{1/2}{1/2} = \frac{2s^2 + 1}{(s^2 + 2)s} = Z_{22} \text{ Verificado } \checkmark$$

5.º) Verificación



Condición primitiva T

$$V_1 = A \cdot V_2 + B (-I_2)$$

$$I_1 = C \cdot V_2 + D (-I_2)$$

$$\frac{1}{A} = \frac{V_2}{V_1} \Big|_{I_2=0} \quad , \quad \frac{1}{C} = \frac{V_2}{I_1} \Big|_{I_2=0}$$

Plantas producto matrices transpuestas

$$\begin{bmatrix} 1 & Z_{C3} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_{C2} R_2 & 1 \end{bmatrix} \begin{bmatrix} 1 & Z_{C1} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+Z_{C3} Y_{C2} R_2 & Z_{C3} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & Z_{C1} \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+Z_{C3} Y_{C2} R_2 + Z_{C3} Z_{C1} & Z_{C3} Z_{C1} \\ Y_{C2} R_2 & 1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

resultado A

$$A = 1 + Z_{C3} Y_{C2} R_2 = 1 + \frac{5}{5} \cdot \frac{1}{25} \cdot 4 = 1 + \frac{4}{25} = \frac{29}{25}$$

$$\frac{1}{C} = \frac{1}{Y_{C2} R_2} = \frac{1}{\frac{1}{5} \cdot 4} = \frac{5}{4}$$

$$\Rightarrow 1 + \frac{5/4}{29/25} = \frac{29 + 5/4}{29} = \frac{116 + 5}{116} = \frac{121}{116} = B$$

resultado B

$$B = (1 + Z_{C3} Y_{C2} R_2) Z_{C1} + Z_{C3} = \frac{29}{25} \cdot \frac{5}{5} + \frac{5}{5} = \frac{29}{25} + 1 = \frac{54}{25}$$

resultado Y22

$$Y_{22} = \frac{1}{B} = \frac{25}{54} = \frac{25}{54} \text{ mhos}$$

resultado Y21

$$Y_{21} = -\frac{1}{B} = -\frac{25}{54} = -\frac{25}{54} \text{ mhos}$$

Exercise #3

Synthesize a quadripole that complies with:

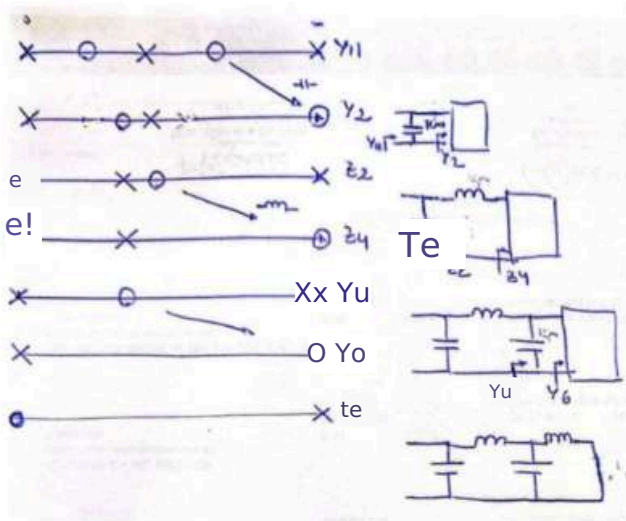
$$Y_{21} = \frac{1}{s(s^2 + 4)}$$

$$Y_{11} = \frac{(s^2 + 1)(s^2 + 9)}{s(s^2 + 4)}$$

$$y_{21} = \frac{1}{s(s^2 + 4)}$$

$$y_{11} = \frac{(s^2 + 1)(s^2 + 9)}{s(s^2 + 4)}$$

$$\frac{I_2}{I_1} = \frac{y_{21}}{y_{11}} = \frac{\frac{1}{s(s^2 + 4)}}{\frac{(s^2 + 1)(s^2 + 9)}{s(s^2 + 4)}} = \frac{1}{(s^2 + 1)(s^2 + 9)}$$



Residue calculation:

$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{y_{11}}{s} = 1$$

$$Y: \quad -s = \frac{s^4 + 10s^2 + 9}{s^3 + 4s} - s = \frac{6s^2 + 9}{s^3 + 4s}$$

$$z_2 = \frac{s^3 + 4s}{6s^2 + 9}$$

$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{z_2}{s} = \lim_{s \rightarrow \infty} \frac{s^2 + 4}{6s^2 + 9} = \frac{1}{6}$$

$$z_4 = z_2 - \frac{1}{6}s = \frac{s^3 + 4s - \frac{1}{6}s(6s^2 + 9)}{6s^2 + 9} = \frac{4s - \frac{3}{2}s}{6s^2 + 9}$$

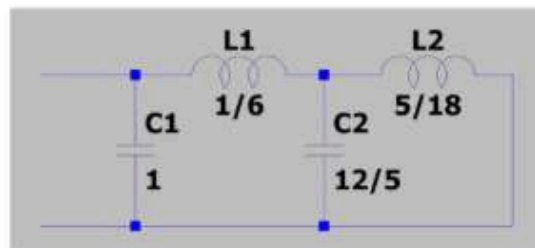
$$z_4 = \frac{5/2 s}{6s^2 + 9}$$

$$y_4 = \frac{6s^2 + 9}{5/2 s}$$

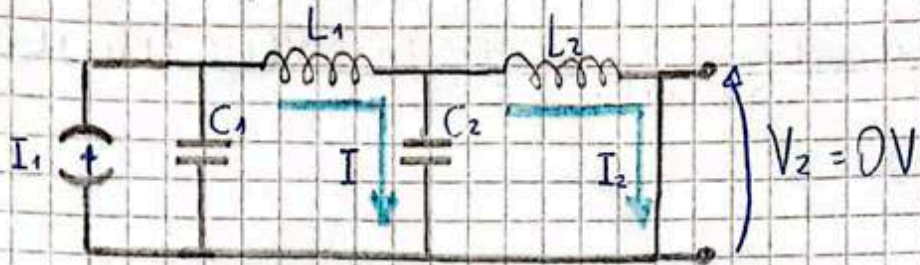
$$k_{\infty} = \lim_{s \rightarrow \infty} \frac{y_4}{s} = \lim_{s \rightarrow \infty} \frac{6s^2 + 9}{5/2 s^2} = \frac{12}{5}$$

$$y_6 = y_4 - \frac{12}{5}s = \frac{6s^2 + 9 - \frac{12}{5}s \frac{5}{2}s}{\frac{5}{2}s} = \frac{18}{5s}$$

$$z_6 = \frac{5}{18}s$$



Verificación ejercicio 3



$$C_1 = 1 \text{ F} \quad ; \quad L_1 = 1/6 \text{ H}$$

$$C_2 = 12/5 \text{ F} \quad ; \quad L_2 = 5/18 \text{ H}$$

Por mallas:

$$I \cdot \left(\frac{1}{5} + \frac{1}{6} s + \frac{5}{125} \right) - \frac{I_1}{5} - \frac{5}{125} I_2 = 0 \quad (A)$$

$$I_2 \left(\frac{5}{125} + \frac{5 \cdot 5}{18} \right) - \frac{5}{125} I = 0 \rightarrow I = I_2 \left(1 + \frac{25}{3} \right) \quad (B)$$

Reemplazo (B) en (A):

$$I_2 \left(1 + \frac{25}{3} \right) \left(\frac{17 + 25s^2}{125} \right) - \frac{I_1}{5} - \frac{5}{125} I_2 = 0$$

$$I_2 \left(\frac{17 + 25s^2}{125} + \frac{(34/3)s^2 + (4/3)s^4}{125} - \frac{5}{125} \right) = \frac{I_1}{5}$$

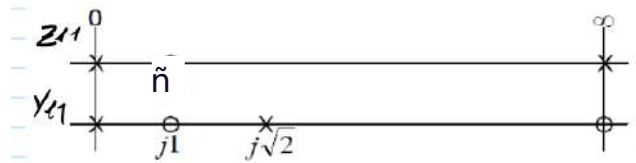
$$I_2 \left(\frac{5s^4 + 10s^2 + 9}{125} \right) = \frac{3}{4} \cdot \frac{I_1}{5}$$

$$\frac{I_2}{I_1} = \frac{9}{(5s^2 + 9)(5s^2 + 1)}$$

Exercise #4

Synthesize as symmetric Lattice:

4) Synthesize as symmetric Lattice

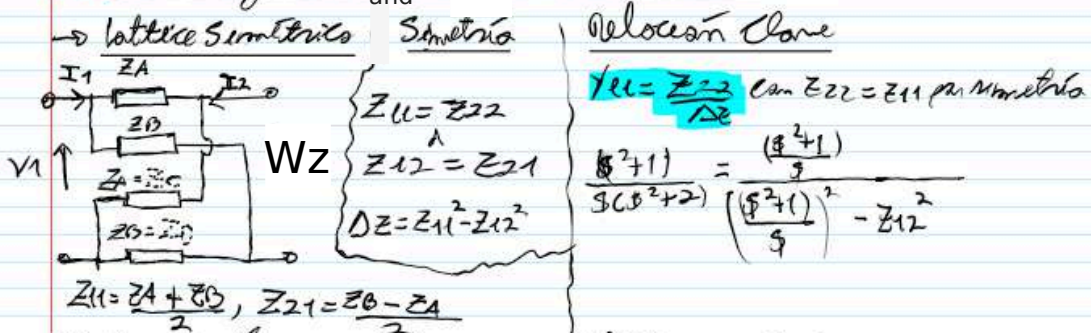


10) Obtenga expresiones de Z_{11} Already in the diagram

→ $Z_{11} = \frac{UN}{4}$, $Y_{11} = \frac{(s^2+1)}{s(s^2+2)}$

Recordemos que los polos de transmisión en $s=0$ no se pasan y que los polos en $s=j\omega$ se pasan.

2.2) Encontrar expresiones de Z_{11} and Z_{22} for a symmetric lattice



3) Introduciendo la expresión queda: $\frac{(s^2+1)}{s(s^2+2)} = \frac{(s^2+1)}{s \left(\frac{(s^2+1)^2}{s^2} - Z_{12}^2 \right)}$ ES

$$\frac{((s^2+1)^2 - Z_{12}^2)s}{s^2(s^2+2)} = 1 \Rightarrow \frac{(s^2+1)^2 - Z_{12}^2}{s^2} = s^2+2 \Rightarrow \frac{(s^2+1) - (s^2+2) - Z_{12}^2}{s^2}$$

$$Z_{12}^2 = \frac{(s^2+1)^2 - s^2(s^2+2)}{s^2} = \frac{s^4+2s^2+1 - s^4 - 2s^2}{s^2} = \frac{1 - 2s^2}{s^2}$$

$$Z_{12}^2 = \frac{1 - 2s^2}{s^2} \Rightarrow Z_{12} = \frac{1}{s} - 2s = Z_{21}$$

De $Z_{21} = \frac{Z_D - Z_A}{2} \Rightarrow 2Z_{21} + Z_A = Z_D$ ①

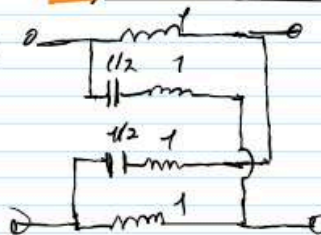
Of $Z_{11} = \frac{Z_A + Z_D}{2} \Rightarrow 2Z_{11} - Z_A = Z_D$ ②

De ① = ② $\Rightarrow 2Z_{21} + Z_A = 2Z_{11} - Z_A$

$$2Z_A = 2Z_{11} - 2Z_{21} \Rightarrow Z_A = Z_{11} - Z_{21}$$

$$Z_A = \frac{(s^2+1)}{s} - \frac{1}{s} = \frac{s^2}{s} = Z_A = \text{Zinductor}$$

Análogamente $Z_D = \frac{1}{s} + 2s = Z_{capacitor} + Z_{inductor}$



Exercise #5

Knowing the transfer:

$$\frac{V_2}{V_1} = \frac{s^2 + 1}{2s^2 + 1}$$

- Obtain a circuit with 3 inductors and 1 capacitor.
- Obtain a circuit with 3 capacitors and 1 inductor.

a)

5) a) $\frac{V_2}{V_1} = \frac{s^2 + 1}{2s^2 + 1} = -\frac{Y_{21}}{Y_{22}}$

Plantamos como polinomio auxiliar de Y_{22} a: $S(S^2, \frac{3}{4})$

$Y_{22} = \frac{2S^2 + 1}{S(S^2, \frac{3}{4})}$

$Z_2 = \frac{1}{Y_{22}} - K_{00} \cdot S$; $K_{00} = \frac{S(S^2, \frac{3}{4})}{S(2S^2 + 1)} \Big|_{S=0} = \frac{1}{4}$

$Z_2 = \frac{S(S^2 + \frac{3}{4})}{2S^2 + 1} - \frac{1}{4} \cdot S = \frac{S(S^2 + \frac{3}{4}) - \frac{1}{4} S(2S^2 + 1)}{2S^2 + 1}$

$Z_2 = \frac{S^3 + \frac{3}{4} S - \frac{1}{2} S^3 - \frac{1}{4} S}{2S^2 + 1} = \frac{\frac{1}{2} S^3 + \frac{1}{2} S}{2S^2 + 1} \rightarrow Y_2 = \frac{2S^2 + 1}{\frac{1}{2} S(S^2 + 1)}$

NOTA

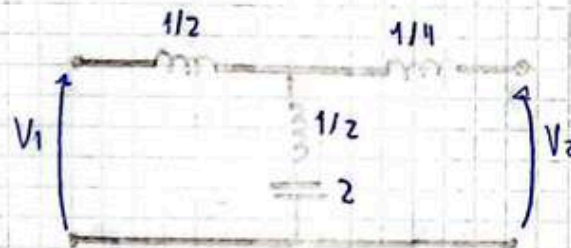
$$Y_4 = Y_2 - \frac{2K_1 S}{S^2 + 1} \quad ; \quad 2K_1 = \lim_{S^2 \rightarrow -1} \frac{(2S^2 + 1) \cdot (S^2 + 1)}{\frac{1}{2}S(S^2 + 1) \cdot S} = 2$$

$$Y_4 = \frac{2S^2 + 1}{\frac{1}{2}S(S^2 + 1)} - \frac{2S}{S^2 + 1} = \frac{2S^2 + 1 - S^2}{\frac{1}{2}S(S^2 + 1)} = \frac{S^2 + 1}{\frac{1}{2}S(S^2 + 1)} = \frac{1}{\frac{1}{2}S}$$

$$Z_4 = \frac{1}{\frac{1}{2}S}$$

$$Z_6 = Z_4 - K_\infty \cdot S \quad ; \quad K_\infty = \lim_{S \rightarrow \infty} \frac{\frac{1}{2}S}{S} = \frac{1}{2}$$

$$Z_6 = \frac{1}{2}S - \frac{1}{2} \cdot S = 0$$



Verificación desde los parámetros impedancia:

$$Z_{12} = Z_{21} = \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + 1}{2S}$$

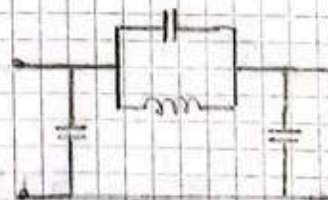
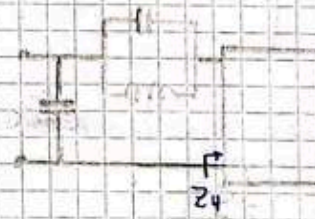
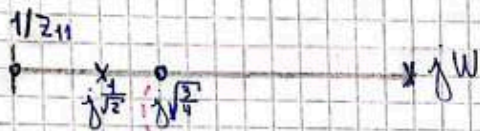
$$Z_{11} = \frac{S}{2} + \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + S^2 + 1}{2S} = \frac{2S^2 + 1}{2S}$$

$$\frac{V_2}{V_1} = \frac{Z_{21}}{Z_{11}} = \frac{\frac{S^2 + 1}{2S}}{\frac{2S^2 + 1}{2S}} = \frac{S^2 + 1}{2S^2 + 1}$$

$$b) \frac{V_2}{V_1} = \frac{S^2+1}{2S^2+1} = \frac{Z_{21}}{Z_{11}}$$

Plantearnos como polinomio auxiliar de Z_{11} a: $S(S^2 + \frac{3}{4})$

$$Z_{11} = \frac{2S^2+1}{S(S^2+\frac{3}{4})}$$



$$Y_2 = \frac{1}{Z_{11}} - K_{\infty}' \cdot S \quad ; \quad K_{\infty}' = \left. \frac{S(S^2 + \frac{3}{4})}{S(2S^2+1)} \right|_{S^2=-1} = \frac{1}{4}$$

$$Y_2 = \frac{S(S^2 + \frac{3}{4})}{2S^2+1} - \frac{1}{4}S = \frac{\frac{1}{2}S(S^2+1)}{2S^2+1} \rightarrow Z_2 = \frac{2S^2+1}{\frac{1}{2}S(S^2+1)}$$

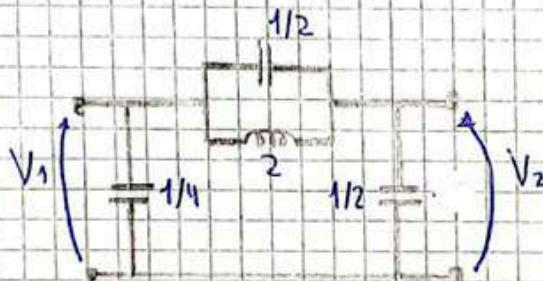
NOTA

$$Z_4 = Z_2 - \frac{2K_1 \cdot S}{S^2 + 1} ; 2K_1 = \lim_{S^2 \rightarrow -1} \frac{(2S^2 + 1)(S^2 + 1)}{\frac{1}{2}S(S^2 + 1)S} = 2$$

$$Z_4 = \frac{2S^2 + 1}{\frac{1}{2}S(S^2 + 1)} - \frac{2S}{S^2 + 1} = \frac{1}{\frac{1}{2}S} \rightarrow Y_4 = \frac{1}{2}S$$

$$Y_6 = Y_4 - K_\infty \cdot S ; K_\infty = \lim_{S \rightarrow \infty} \frac{\frac{1}{2}S}{S} = \frac{1}{2}$$

$$Y_6 = \frac{1}{2}S - \frac{1}{2}S = 0$$



• Verificación desde los parámetros admitancia:

$$-Y_{21} = -Y_{12} = \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + 1}{2S}$$

$$Y_{22} = \frac{S}{2} + \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + S^2 + 1}{2S} = \frac{2S^2 + 1}{2S}$$

$$\frac{V_2}{V_1} = \frac{-Y_{21}}{Y_{22}} = \frac{\frac{S^2 + 1}{2S}}{\frac{2S^2 + 1}{2S}} = \frac{S^2 + 1}{2S^2 + 1}$$

Exercise #6

Synthesize a quadripole that complies with:

$$Y_{11} = \frac{3s(s'' + 7/3)}{(s^2 + 2)(s^2 + 5)}$$

$$Y_{21} = \frac{s(s^2 + 1)}{(s^2 + 2)(s^2 + 5)}$$

[illegible]

$$Y_6(s) = Y_4(s) - \frac{2K_1 s}{(s^2+1)^2}, \quad 2K_1 = \lim_{s \rightarrow -1} Y_4(s) \cdot \frac{(s^2+1)^2}{s} = \lim_{s \rightarrow -1} \frac{3s(s^2+7)}{(s^2+3)} \cdot \frac{(s^2+1)^2}{s}$$

$$2K_1 = \lim_{s \rightarrow -1} \frac{3(s^2+7)}{(s^2+3)} = \frac{3(-1+7)}{-1+3} = 2 = 2K_1$$

$$Y_6(s) = \frac{3s(s^2+7)}{(s^2+1)(s^2+3)} - \frac{2s}{(s^2+1)} = \frac{3s(s^2+7) - 2s(s^2+3)}{(s^2+1)(s^2+3)} = \frac{3s^3 + 7s - 2s^3 - 6s}{(s^2+1)(s^2+3)}$$

$$Y_6(s) = \frac{s(s^2+1)}{(s^2+1)(s^2+3)} = \frac{s}{(s^2+3)} = Y_6(s), \quad \frac{s^2+3}{s} = 36(s)$$

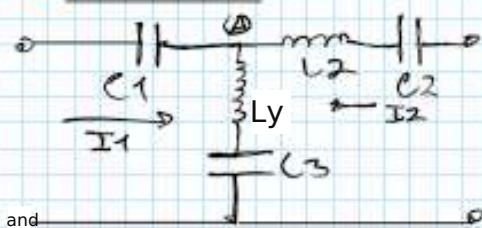
$$Z_8(s) = Z_6(s) - K_{\infty} s, \quad K_{\infty} = \lim_{s \rightarrow \infty} \frac{Z_6(s)}{s} = \lim_{s \rightarrow \infty} \frac{s^2+3}{s} \cdot \frac{1}{s} = 1 = K_{\infty}$$

$$Z_8(s) = \frac{s^2+3}{s} - 1 \cdot s = \frac{s^2+3-s^2}{s} = \frac{3}{s} = Z_8(s)$$

$$Z_{10}(s) = Z_8(s) - \frac{K_0}{s}, \quad K_0 = \lim_{s \rightarrow 0} Z_8(s) \cdot s = \lim_{s \rightarrow 0} \frac{3}{s} \cdot s = 3 = K_0$$

$$Z_{10}(s) = \frac{3}{s} - \frac{3}{s} = 0 \Rightarrow \text{cable}$$

32) Circuits



and

$$\text{and } z = \frac{1}{\frac{1}{Cs} + sL_3} = \frac{1}{\frac{s}{Cs} + sL_3} = \frac{Cs}{s^2 L_3 C + 1}$$

$$Y_{L_3, C_3} = \frac{sCs}{s^2 L_3 C + 1} \cdot \frac{1/(sL_2)}{1/(sL_2) + 1/(Cs)} = \frac{s(1/L_2)}{s^2 + \frac{1}{L_2 C}}$$

$$\frac{2K_1 s}{s^2+4} \Rightarrow \begin{cases} \frac{L_2}{L_3} = 2 \\ \frac{1}{L_3 C_3} = 1 \end{cases} \Rightarrow \begin{cases} L_2 = 2L_3 \\ \frac{1}{(1/2)L_3} = 1 \Rightarrow 2 = C_3 \end{cases}$$

$$C_1 = \frac{1}{K_0} = 1, \quad C_2 = \frac{1}{K_0} = 1/3, \quad L_2 = K_0 = 1$$

33) Verification

Recall that $-\frac{I_2}{I_1} \Big|_{V_2=0} = T(s) = -Y_{21}$, But we can obtain something to $Y(s)$

And the

$$\begin{bmatrix} 1 & Z_e \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ Y_{12} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & Z_{L_2} \\ 0 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 + Z_e Y_{12} & 1 \cdot 0 + Z_e \\ 0 \cdot 1 + Y_{12} Z_{L_2} & 0 \cdot 0 + 1 \cdot 1 \end{bmatrix} \begin{bmatrix} 1 & Z_{L_2} C_2 \\ 0 & 1 \end{bmatrix}$$

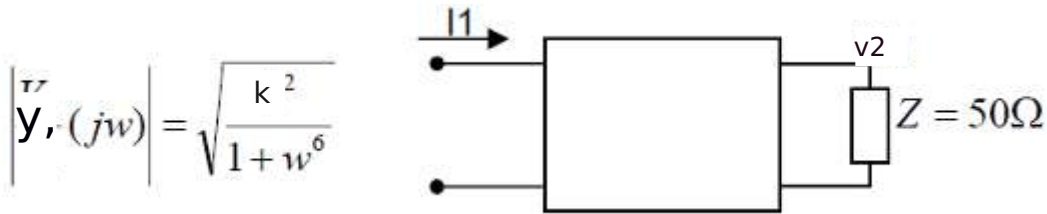
$$[T] \Rightarrow \begin{bmatrix} 1 + Z_e Y_{12} C_2 + 0 & (1 + Z_e Y_{L_2} C_2) \cdot Z_{L_2} C_2 + Z_e \\ Y_{L_2} C_2 \cdot 1 + 1 \cdot 0 & Y_{L_2} C_2 \cdot Z_{L_2} C_2 + 1 \cdot 1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

$$D = Y_{L_2} C_2 \cdot Z_{L_2} C_2 + 1 = \frac{s(1/L_2)}{s^2 + \frac{1}{L_2 C_2}} \cdot \frac{1}{s} + 1 = \frac{s \cdot 2}{s^2 + 1} \cdot \left(\frac{3}{s} + 1 \right) + 1 = \frac{2s}{s^2 + 1} \cdot \frac{3 + s^2}{s} + 1 =$$

$$D = \frac{2(3 + s^2)(s^2 + 1)}{s^2 + 1} = \frac{2(3 + s^2)}{s^2 + 1} = \frac{3(s^2 + 7)}{s^2 + 1} \Rightarrow \frac{1}{D} = \frac{(s^2 + 1)}{3(s^2 + 7)} = -Y_{21} = T(s) \checkmark$$

Exercise #7

Synthesize the following non-dissipative transfer:



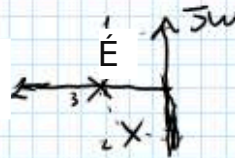
1) Verifiquemos que es la transferencia de 11m Butter
Po = 1000 W.

Mr $g = 5W \Rightarrow \frac{g}{5}$, $Y_{01} |H(w)|^2 = \frac{1}{1+w^2m}$, $g=1$ Componentes
Máximo Planificado

Por Le Y $w \Rightarrow | \frac{V_2(s)}{V_1(s)} | = \sqrt{\frac{K^2}{1+w^6}} \Rightarrow |H(w)|^2 = \frac{K^2}{1+w^6}$

$\Rightarrow$ Butter of orden $m=3$.

$H(s) = \frac{K}{s^3 + 1.5s + 1}$ A ca



$q_{1,2} = \frac{1}{2 \cos(\pi/2)}$, $q_3 = 1$, $w_0 = 1$

Asomados $\frac{V_2(s)}{V_1(s)} = H(s) = \frac{K^2}{s^3 + s^2 + s + 1} = \frac{K^2}{s^3 + 2s^2 + 2s + 1}$

Por ser Transferencia $\Rightarrow$ Tiene que existir mismo control de polos
y ceros $\Rightarrow \arg(den(s)) = \arg(num(s)) = Y$ Vamos por inspección
que hay 3 polos y fre deben existir 3 ceros en el as that mane
en la expresión.

2) Queremos igualar la expresión $\frac{V_2}{V_1}$ con la del cuadrupolo.

$\Rightarrow \frac{V_2}{V_1} = \frac{1}{A + B \cdot \frac{V_2}{V_1} + Z_0} \Rightarrow Z_0 = 0$
D $Z_L = 50 \Omega$

$\Rightarrow \frac{V_2}{V_1} = \frac{1}{Alex} - \text{donde } A = \frac{-Y_{22}}{Y_{21}}, B = \frac{1}{-Y_{21}}$
 $Y_L = Ls$

$\Rightarrow \frac{V_2}{V_1} = \frac{1}{\frac{-Y_{22}}{Y_{21}} + \frac{1}{-Y_{21}} \cdot Y_L} = \frac{-Y_{21}}{Y_{22} + Y_L} = \frac{K^2}{s^3 + 2s^2 + 2s + 1}$

$$\frac{V_2}{V_1} = \frac{1}{\frac{-Y_{22} + 1}{Y_{21}} \cdot \bar{Y}_L} = \frac{-Y_{21}}{Y_{22} + Y_L} = \frac{A^2}{s^3 + 2s^2 + 2s + 1}$$

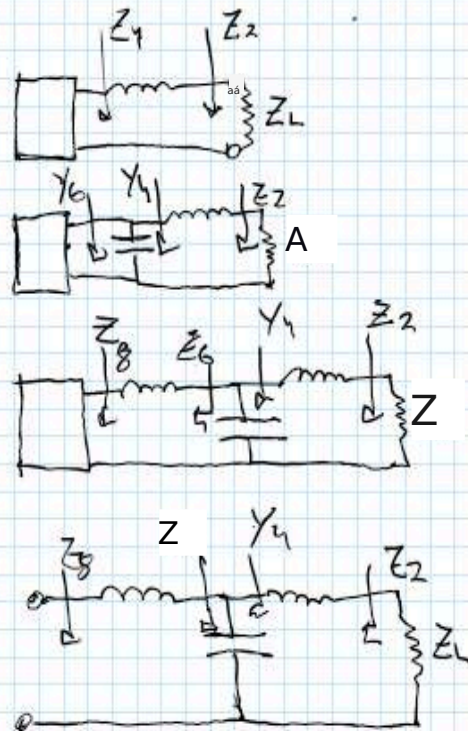
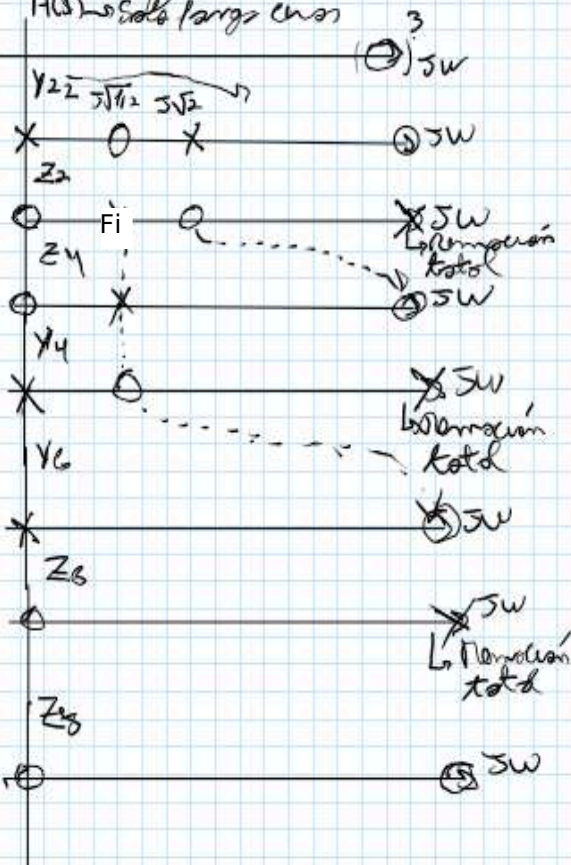
32) Ahora se divide por h_o para el valor del denominador de 4(8) e se demuestra que hay $month$ soluciones

$$\Rightarrow \frac{\frac{K^2}{s^3 + 2s}}{\frac{s^3 + 2s^2 + 2s + 1}{s^3 + 2s}} = \frac{K^2}{\frac{s^3 + 2s}{2s^2 + 1} + 1}$$

$\frac{K^2}{s^3 + 2s} = -Y_{21}$
 $\frac{2s^2 + 1}{s^3 + 2s} = \text{And to}$
 $\text{And } (s^3 + 2s) = 1$

42) Sustituimos Y_{22} que es función de excitación, a partir del método gráfico

H(3) solo para caso



6) Cálculo de residuos

$$Y_{22} = \frac{2s^2 + 1}{s^3 + 2s}, \quad Z_2 = \frac{1}{Y_{22}} = \frac{s^3 + 2s}{2s^2 + 1} \checkmark$$

$$\bullet Z_4(s) = Z_2(s) - K_{0,1} \cdot s, \quad K_{0,1} = \lim_{s \rightarrow \infty} \frac{Z_2(s)}{s} = \lim_{s \rightarrow \infty} \frac{s^3 + 2s}{2s^3 + 1} = \frac{1}{2}$$

$$K_{0,1} = \frac{1}{2}$$

$$\bullet Z_4(s) = \frac{s^3 + 2s}{2s^2 + 1} - \frac{1}{2} \cdot s = \frac{s^3 + 2s - s^2 - \frac{1}{2}s}{2s^2 + 1} = \frac{(3/2)s}{2s^2 + 1} = Z_4(s)$$

$$\frac{2s^2 + 1}{(3/2)s} = Y_4(s)$$

$$\bullet Y_6(s) = Y_4(s) - K_{0,2} \cdot s, \quad K_{0,2} = \lim_{s \rightarrow \infty} \frac{Y_4(s)}{s} = \lim_{s \rightarrow \infty} \frac{2s^2 + 1}{(3/2)s} \cdot \frac{1}{s} = \frac{2}{3/2} = 4/3$$

$$K_{0,2} = 4/3,$$

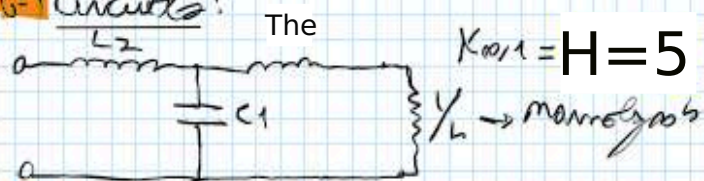
$$\bullet Y_6(s) = \frac{2s^2 + 1}{(3/2)s} - (4/3) \cdot s = \frac{2s^2 + 1 - 2s^2}{(3/2)s} = \frac{1}{(3/2)s} = Y_6(s), \quad (3/2)s = Z_6(s)$$

$$\bullet Z_8(s) = Z_6(s) - K_{0,3} \cdot s, \quad K_{0,3} = \lim_{s \rightarrow \infty} \frac{Z_6(s)}{s} = \frac{(3/2)s}{s} = 3/2$$

$$K_{0,3} = 3/2$$

$$\bullet Z_8(s) = (3/2)s - (3/2)s = 0 \rightarrow \text{cable}$$

6.1) Circuito:



7.2) Verificación

$$\begin{bmatrix} 1 & Z_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_{c1} & 1 \end{bmatrix} \begin{bmatrix} 1 & Z_1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ Y_L & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 + Z_2 \cdot Y_{c1} & Z_2 \\ Y_{c1} & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & Z_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ Y_L & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 + Z_2 \cdot Y_{c1} & (1 + Z_2 \cdot Y_{c1})Z_1 + Z_2 \\ Y_{c1} & Y_{c1} \cdot Z_1 + 1 \end{bmatrix} \begin{bmatrix} 1 & e \\ Y_L & 1 \end{bmatrix} =$$

Yes

$$\begin{bmatrix} 1+Z_{L2}Y_{C1} & (1+Z_{L2}Y_{C1})Z_{L1}+Z_{L2} \\ Y_{C1}Z_{L1}+1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ Y_L \end{bmatrix} L =$$

$$\begin{bmatrix} (1+Z_{L2}Y_{C1})(1+Z_{L2}Y_{C1})Z_{L1}+Z_{L2} & Z_{L2}Y_{C1}Z_{L1}+Z_{L2} \\ Y_{C1}Z_{L1}+1 & 1 \end{bmatrix} \begin{bmatrix} V_2 \\ Y_L \end{bmatrix}$$

$$A' \Rightarrow \frac{V_2}{V_1} = \frac{1}{A'}$$

$$\frac{V_2}{V_1} = \frac{1}{1+(3/2)\Omega \cdot (4/3)\Omega + ((1+(3/2)\Omega \cdot (4/3)\Omega) \cdot (1/2)\Omega + (3/2)\Omega) \cdot 1}$$

$$\frac{V_2}{V_1} = \frac{1}{1+2\Omega^2 + ((1+2\Omega^2) \cdot (1/2)\Omega + (3/2)\Omega)}$$

$$\frac{V_2}{V_1} = \frac{1}{1+2\Omega^2 + (\frac{1}{2}\Omega + \Omega^3 + \frac{3}{2}\Omega)}$$

$$\text{goes} = \frac{1}{1+2\Omega^2 + 2\Omega + \Omega^3}$$

$$\frac{V_2}{V_1} = \frac{A}{\Omega^3 + 2\Omega^2 + 2\Omega + 1} \text{ se demuestra! } qa \quad K=1 \text{ It Normalizado}$$

Desarmamos la giradora $Z_N = 50\Omega = \text{EL del enunciado}$

$$L_1 = L_1 N \cdot Z_N = \frac{1}{2} \cdot 50\Omega = 25 \text{ Hy}$$

$$L_2 = L_2 N \cdot Z_N = \frac{3}{2} \cdot 50\Omega = \frac{150}{2} = 75 \text{ Hy}$$

$$C_1 = C_1' \cdot Z_N = \frac{4}{3} \cdot 50\Omega = \frac{200}{3} \text{ And}$$

$$N = N \cdot Z_N = 1 \cdot 50\Omega = 50\Omega$$

se demuestra tag que se reemplazamos el sistema por el que antes
 $\Rightarrow$ obtenemos $\frac{V_2}{V_1}$ desarmado $\Rightarrow \frac{V_2}{V_1} = \frac{\frac{1}{2500}}{\Omega^3 + 2\Omega^2 + \frac{1}{250}\Omega + \frac{1}{2500}}, K^2 = \frac{1}{2500}$

Exercise #8

Synthesize the following RC quadripole:

$$-\frac{I_2}{I_1} = H \cdot \frac{(s^2 + 5s + 4)}{(s^2 + 8s + 12)}$$

$$Z_{21} = 6H$$



Comenzamos igualando la expresión $-\frac{I_2}{I_1}$ o la del quadripole

$$-\frac{I_2}{I_1} = \frac{1}{A \cdot Y_g \cdot Z_L + C \cdot Z_L + Y_g \cdot D + D}, \text{ donde por simple inspección queda } Y_g = 0, Z_L = 1$$

$$-\frac{I_2}{I_1} = \frac{1}{C \cdot Z_L + D}, C = \frac{1}{Z_{21}}, D = \frac{Z_{22}}{Z_{21}}$$

$$-\frac{I_2}{I_1} = \frac{1}{\frac{1}{Z_{21}} \cdot Z_L + \frac{Z_{22}}{Z_{21}}} \cdot \frac{Z_{21}}{Z_{21}} = \frac{Z_{21}}{Z_L + Z_{22}} = \frac{Z_{21}}{1 + Z_{22}}$$

$$\boxed{-\frac{I_2}{I_1} = \frac{Z_{21}}{1 + Z_{22}}} \Rightarrow \text{Igualando} \Rightarrow \frac{Z_{21}}{1 + Z_{22}} = H \cdot \frac{s^2 + 5s + 4}{(s^2 + 8s + 12)}$$

$$\Rightarrow \frac{6H}{1 + Z_{22}} = H \cdot \frac{(s^2 + 5s + 4)}{s^2 + 8s + 12} \Rightarrow 6 \frac{(s^2 + 8s + 12)}{s^2 + 5s + 4} = (1 + Z_{22})$$

$$\Rightarrow 6 \cdot \frac{(s^2 + 8s + 12)}{s^2 + 5s + 4} = 1 + Z_{22}$$

$$\Rightarrow \frac{6(s^2 + 8s + 12)}{s^2 + 5s + 4} - 1 = Z_{22}$$

$$\Rightarrow \frac{6s^2 + 48s + 72 - s^2 - 5s - 4}{s^2 + 5s + 4} = Z_{22}$$

$$\Rightarrow \boxed{\frac{5s^2 + 43s + 68}{s^2 + 5s + 4} = Z_{22}} \Rightarrow \frac{5(s^2 + 8.6s + 13.6)}{(s+1)(s+4)} = \frac{5 \cdot (s+2.1)(s+6.5)}{(s+1)(s+4)}$$

$$Z_2(s) = \frac{5 \cdot (s+5.6)}{(s+4)}, \quad Z_2(0) = \frac{5(0+5.6)}{0+4} = 7$$

$$Z_4(s) = Z_2(s) - \frac{2K_2}{s+4}, \quad Z_4(s) \Big|_{s=-4} = 0 = \left(Z_2(s) - \frac{2K_2}{(s+4)} \right) \Big|_{s=-4}$$

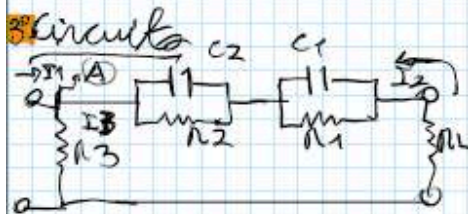
$$\frac{5(s+5.6)}{(s+4)} - \frac{2K_2}{(s+4)} = 0 \Rightarrow \frac{5(s+5.6)}{(s+4)} = \frac{2K_2}{(s+4)} \Rightarrow 5 \cdot (s+5.6) = 2K_2$$

$$\Rightarrow 5(-4+5.6) = 2K_2 \Rightarrow 2K_2 = 8$$

$$Z_4(s) = Z_2(s) - \frac{8}{s+4} = \frac{5(s+5.6)}{(s+4)} - \frac{8}{(s+4)} = \frac{5s+28-8}{s+4} = \frac{5s+20}{s+4} = 5$$

$$Z_4(s) = 5, \quad Y_4(s) = \frac{1}{5}$$

$$Y_0(s) = Y_4(s) - \frac{1}{5} = 0$$



$$Z_{C1R1} = \frac{2K}{s+5}$$

$$Z_{C1R1} = \frac{1}{Y_{C1R1}} = \frac{1}{\frac{1}{s+5}} \cdot \frac{1/C}{1/C} = \frac{1/C}{s + \frac{1}{RC}}$$

$$\Rightarrow Z_{C1R1} = \frac{1/C1}{s + \frac{1}{C1R1}} \Rightarrow \begin{cases} 1/C1 = 2K1 = 1 \\ \frac{1}{C1R1} = 5 \end{cases}$$

$$\begin{cases} C1 = \frac{1}{5} F \\ R1 = 1 \end{cases}$$

$$\Rightarrow Z_{C2R2} = \frac{1/C2}{s + \frac{1}{C2R2}} \Rightarrow \begin{cases} \frac{1}{C2} = 2K2 = 7 \\ C2 = \frac{7}{8} F \\ \frac{1}{C2R2} = 4 \end{cases}$$

$$\Rightarrow R3 = R1 = 1$$

4) Verificación

Desarmamos el circuito, aplicamos Kirchhoff ley de voltajes:

$$V_{R3} = V_{R1} + V_{C1R1} + V_{C2R2} \Rightarrow \text{modo (A)} \quad I_1 + (-I_2) = I_3, \quad Z' = Z_{C1R1} + Z_{C2R2}$$

$$Z' = \frac{1/C1}{s + \frac{1}{C1R1}} + \frac{1/C2}{s + \frac{1}{C2R2}} = \frac{1}{s+1} + \frac{7}{s+4} = \frac{1(s+4) + 7(s+1)}{(s+1)(s+4)} = \frac{1s+4+7s+7}{s^2+5s+4}$$

$$Z' = \frac{1s+11}{s^2+5s+4} = \frac{1s}{s^2+5s+4} + \frac{11}{s^2+5s+4}, \quad (I_1 + (-I_2))R3 = I_2 \cdot Z' + I_2 \cdot R1$$

$$I_1 \cdot R3 = I_2 \cdot Z' + I_2 \cdot R1$$

$$(I_1 + I_2)R_3 = I_2 Z' + I_2 R_L \Rightarrow I_1 R_3 = I_2 Z' + I_2 R_L + I_2 R_3$$

$$\Rightarrow \frac{R_3}{Z' + R_L + R_3} = \frac{I_2}{I_1}$$

$$\Rightarrow \frac{5}{18(s + \frac{5}{18}) + 11.5} \cdot (s^2 + 5s + 4) = \frac{E}{I_1}$$

$$\Rightarrow \frac{5 \cdot (s^2 + 5s + 4)}{18s + 5 + 6(s^2 + 5s + 4)} = \frac{5(s^2 + 5s + 4)}{6s^2 + 48s + 75} = \frac{5(s^2 + 5s + 4)}{6(s^2 + 8s + 12)} = \frac{I_2}{I_1}$$

• Querido verificado y se deduce que $4 = 5/6$

Simulation:

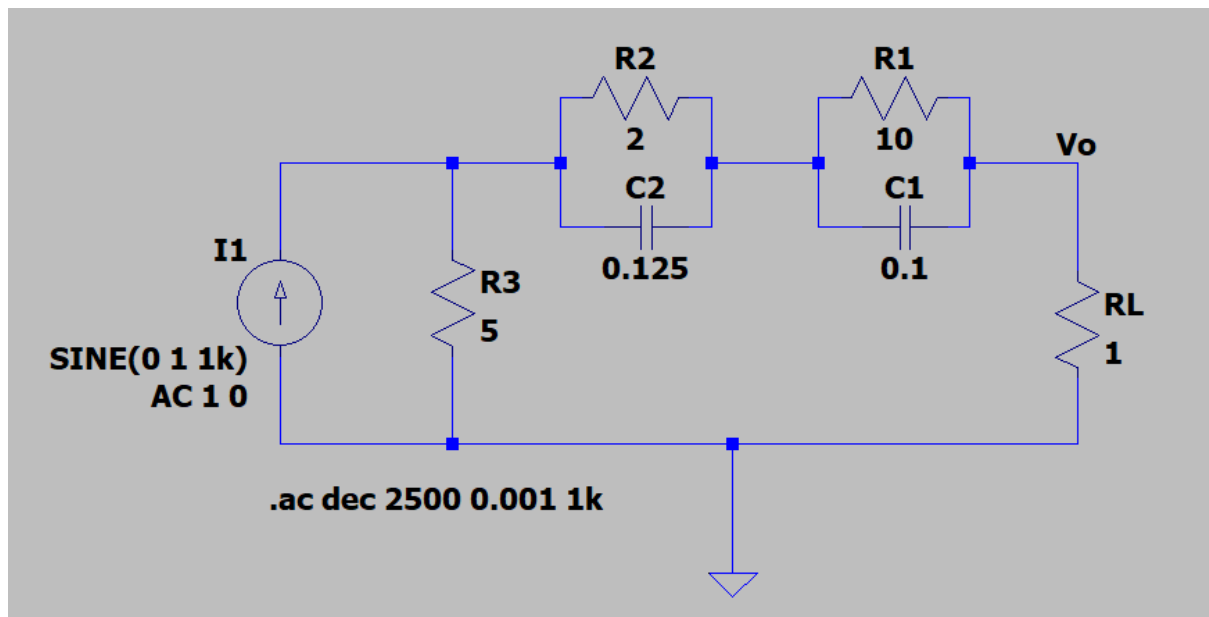
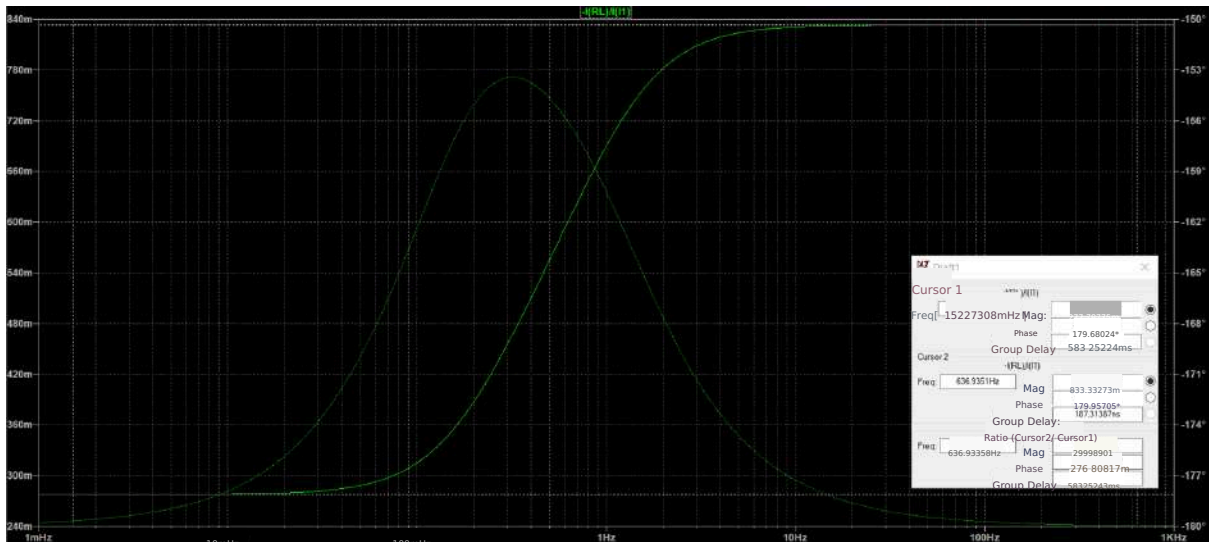


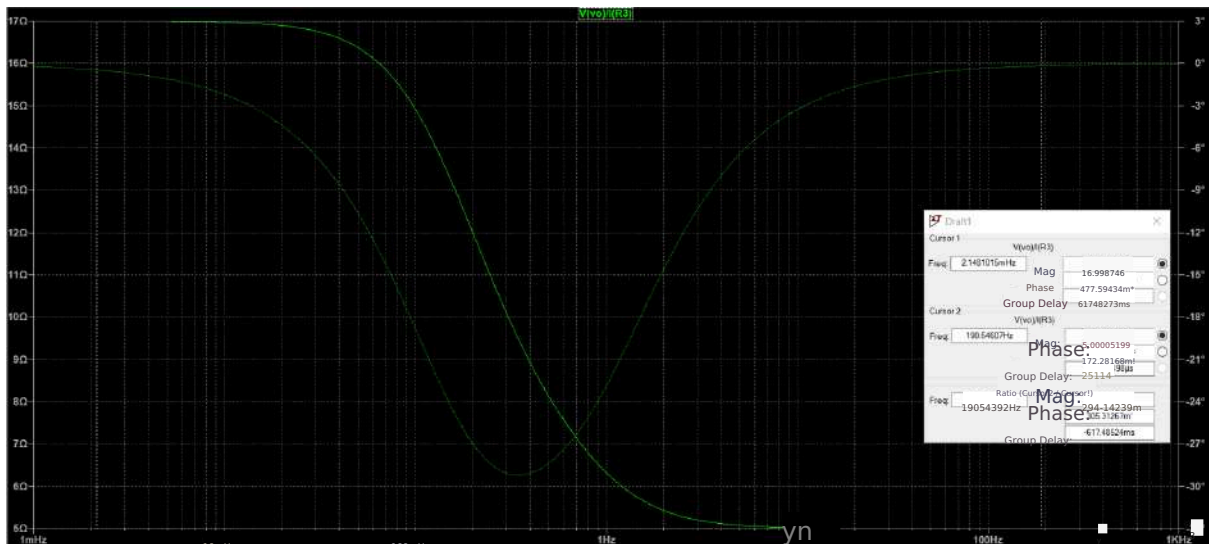
Gráfico 1

Circuit of exercise 8.



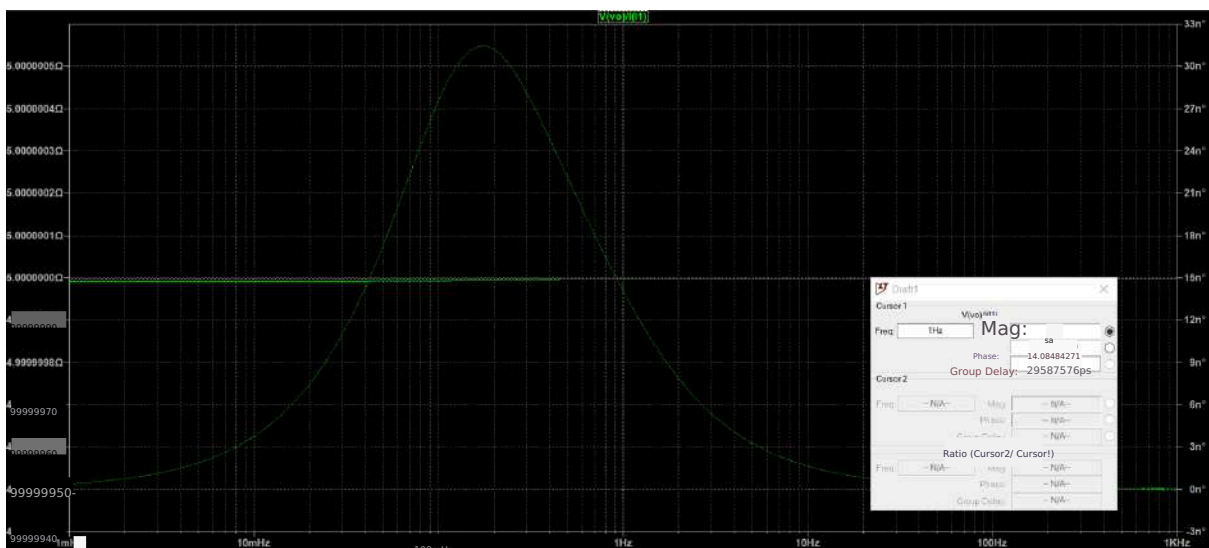
Graph 2

Frequency response of the current transfer.



Graph 3

Frequency response of Z22.



Graph 4

Frequency response of Z21.

Exercise #9

Synthesize using a ladder circuit:

$$T(s) = \frac{(s^2 + 4)(s^2 + 9)}{(s^2 + 1)(s^2 + 3)}$$

9) sintetizar un circuito escalera

$T(s) = \frac{(s^2+4)(s^2+5)}{(s^2+1)(s^2+2)} \Rightarrow T = \frac{V_2}{V_1} = \frac{Z_{21}}{Z_{12}} = \frac{X}{Y} \cdot \frac{e}{e} = \frac{X}{Y}$

$= \frac{(s^2+4)(s^2+5)}{(s^2+1)(s^2+2)}$

Tenemos que usar las funciones de excitación que se pueden dar como sintetizables

N The damos los polos de T

recomendamos de forma parcial

to the

The

recomendamos el polo en admittance

TILA

recomendamos el polo en admittance

$\Rightarrow Z_2 = Z_{11} - K \frac{s}{s^2+4}$

$Z_2 = \frac{Z_{11}}{s^2+4} - K \frac{s}{s^2+4} = \frac{Z_{11} - Ks}{s^2+4}$

$z_h = \frac{(s^2+4)(s^2+5)}{(s^2+1)(s^2+2)} - \frac{2s}{s^2+4}$

$\Rightarrow Y_2 = \frac{s(s^2+2)}{(s^2+1)(s^2+4)}$

$\Rightarrow Y_4 = Y_2 - K \frac{s}{s^2+4}$

$Y_4 = \frac{s(s^2+2)}{(s^2+1)(s^2+4)} - \frac{2s}{s^2+4}$

$= \frac{s(s^2+2)}{(s^2+1)(s^2+4)} - \frac{2s}{s^2+4}$

$= \frac{s(s^2+2) - 2s(s^2+1)}{(s^2+1)(s^2+4)}$

$= \frac{s(s^2+2 - 2s^2 - 2)}{(s^2+1)(s^2+4)}$

$= \frac{s(-s^2)}{(s^2+1)(s^2+4)}$

$= -\frac{s}{(s^2+1)(s^2+4)}$

skycolor

$$\rightarrow z_4 = \frac{\frac{2}{3}(s^2 + \frac{5}{3})(s^2 + 4)}{\frac{2}{3}s^3 + \frac{8}{3}s} = \frac{\frac{2}{3}(s^2 + \frac{5}{3})(s^2 + 4)}{\frac{2}{3}s(s^2 + 4)}$$

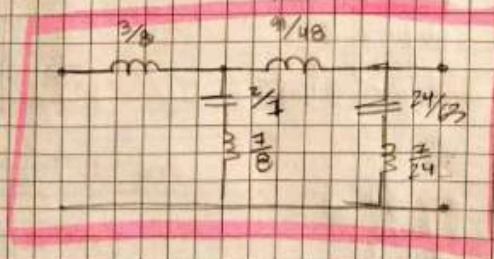
$$\rightarrow z_6 = z_4 - k' \cdot \frac{1}{s} \text{ on } z_4 \Big|_{s=-9} - k' \cdot \frac{1}{s} \Big|_{s=-9} = 0 \Rightarrow z_4 \Big|_{s=-9} = k' \cdot \frac{1}{s} \Big|_{s=-9}$$

$$\text{on } k' = \frac{z_4}{\frac{1}{s}} \Big|_{s=-9} = \frac{\frac{2}{3}(s^2 + \frac{5}{3})(s^2 + 4)}{\frac{2}{3}s(s^2 + 4)} \cdot \frac{1}{s} \Big|_{s=-9} = \frac{\frac{2}{3}(9 + \frac{5}{3})}{\frac{2}{3}(-9)} = \frac{\frac{2}{3} \cdot \frac{32}{3}}{\frac{2}{3}(-9)} = \frac{32}{-9}$$

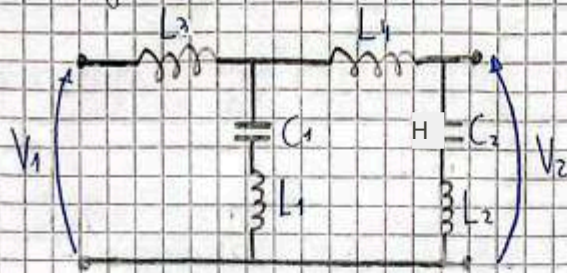
$$\rightarrow z_6 = \frac{\frac{2}{3}(s^2 + \frac{5}{3})}{\frac{2}{3}s} - \frac{32}{9} \cdot \frac{1}{s} = \frac{\frac{2}{3}(s^2 + \frac{5}{3}) - \frac{32}{9}s}{\frac{2}{3}s} = \frac{\frac{1}{3}s^2 + \frac{10}{9} - \frac{64}{9}}{\frac{2}{3}s} = \frac{\frac{1}{3}s^2 - \frac{54}{9}}{\frac{2}{3}s} = \frac{\frac{1}{3}(s^2 - 18)}{\frac{2}{3}s}$$

$$\rightarrow y_6 = \frac{\frac{2}{3}s}{\frac{1}{3}(s^2 - 18)} \rightarrow y_6 = \frac{2}{s^2 - 18} \cdot \frac{1}{s} \text{ on } k = \lim_{s \rightarrow 9} \frac{s^2 - 9}{s^2 - 18} \cdot \frac{\frac{2}{3}s}{\frac{1}{3}(s^2 - 18)} = \frac{24}{-9}$$

$$y_6 = \frac{\frac{2}{3}s}{\frac{1}{3}(s^2 - 18)} - \frac{\frac{24}{9}s}{(s^2 - 18)} = \frac{\frac{2}{3}s}{\frac{1}{3}(s^2 - 18)} - \frac{\frac{8}{3}s}{(s^2 - 18)} = \frac{\frac{2}{3}s - \frac{8}{3}s}{\frac{1}{3}(s^2 - 18)} = \frac{-\frac{6}{3}s}{\frac{1}{3}(s^2 - 18)} = \frac{-2s}{s^2 - 18} = 0$$



Verificación ejercicio 9



$$C_1 = 2/7 \text{ F} ; C_2 = 24/63 \text{ F} ; L_3 = 3/8 \text{ H} \\ L_1 = 7/8 \text{ H} ; L_2 = 7/24 \text{ H} ; L_4 = 9/48 \text{ H}$$

$$\begin{pmatrix} 1 + \frac{SL_3}{SL_1 + 1} & SL_3 \\ \frac{1}{SL_1 + 1} & 1 \end{pmatrix} \begin{pmatrix} 1 + \frac{SL_4}{SL_2 + 1} & SL_4 \\ \frac{1}{SL_2 + 1} & 1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} = [T]$$

Calculo el parámetro A para obtener luego la transferencia de tensión deseada.

$$A = \left(1 + \frac{SL_3}{SL_1 + 1} \right) \left(1 + \frac{SL_4}{SL_2 + 1} \right) + \frac{SL_3}{SL_2 + 1}$$

$$A = \left(\frac{S^2 L_1 C_1 + S^2 L_3 C_1 + 1}{S^2 L_1 C_1 + 1} \right) \left(\frac{S^2 L_2 C_2 + S^2 L_4 C_2 + 1}{S^2 L_2 C_2 + 1} \right) + \frac{S^2 L_3 C_2}{S^2 L_2 C_2 + 1}$$

$$A = \frac{(S^2 L_1 C_1 + S^2 L_3 C_1 + 1)(S^2 L_2 C_2 + S^2 L_4 C_2 + 1) + (S^2 L_3 C_2) \cdot (S^2 L_2 C_2 + 1)}{(S^2 L_1 C_1 + 1)(S^2 L_2 C_2 + 1)}$$

$$A = \frac{S^4 L_1 C_1 L_2 C_2 + S^4 L_1 C_1 L_4 C_2 + S^4 L_3 C_1 L_2 C_2 + S^4 L_3 C_1 L_4 C_2 + S^2 L_3 C_2 L_2 C_2 + S^2 L_3 C_2 + S^2 L_2 C_2 + S^2 L_4 C_2 + S^2 L_1 C_1 + S^2 L_3 C_1 + 1}{(S^2 L_1 C_1 + 1)(S^2 L_2 C_2 + 1)}$$

NOTA

$$A = \frac{S^4(L_1C_1L_2C_2 + L_1C_1L_3C_2 + L_3C_1L_2C_2 + L_3C_1L_4C_2 + L_3C_2L_4C_1) + S^2(L_3C_1 + L_2C_2 + L_4C_1 + L_3C_1) + 1}{(S^2L_1C_1 + 1)(S^2L_2C_2 + 1)}$$

Reemplazo valores:

$$A = \frac{S^4\left(\frac{7}{8} \cdot \frac{2}{7} \cdot \frac{7}{24} \cdot \frac{24}{63} + \frac{7}{8} \cdot \frac{2}{7} \cdot \frac{91}{48} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{7}{24} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{2}{7} \cdot \frac{91}{48} \cdot \frac{24}{63} + \frac{3}{8} \cdot \frac{24}{63} \cdot \frac{7}{8} \cdot \frac{2}{7}\right) + S^2\left(\frac{3}{8} \cdot \frac{24}{63} + \frac{7}{24} \cdot \frac{24}{63} + \frac{91}{48} \cdot \frac{24}{63} + \frac{7}{8} \cdot \frac{2}{7} + \frac{3}{8} \cdot \frac{2}{7}\right) + 1}{\left(S^2 \cdot \frac{7}{8} \cdot \frac{2}{7} + 1\right)\left(S^2 \cdot \frac{7}{24} \cdot \frac{24}{63} + 1\right)}$$

$$A = \frac{S^4 \frac{4}{3} + S^2 \frac{4}{3} + 1}{(S^2 \frac{4}{3} + 1)(S^2 \frac{4}{3} + 1)} = \frac{\frac{1}{3}(S^4 + S^2 \cdot 4 + 3)}{\frac{4}{3} \cdot \frac{4}{3}(S^2 + 4)(S^2 + 9)}$$

$$A = \frac{12(S^2 + 1)(S^2 + 3)}{(S^2 + 4)(S^2 + 9)}$$

Donde finalmente la transferencia de tensión:

$$A^{-1} = \frac{V_2}{V_1} = \frac{1}{12} \frac{(S^2 + 4)(S^2 + 9)}{(S^2 + 1)(S^2 + 3)}$$

Simulation:

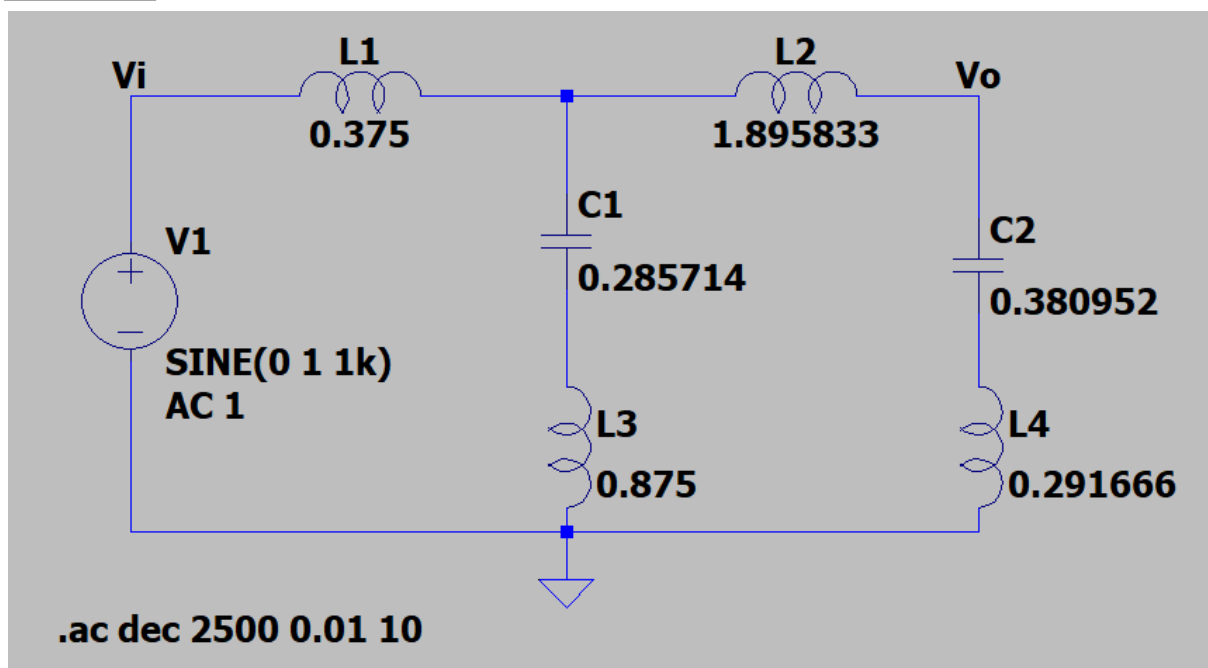


Gráfico 5

Circuit of exercise 9.

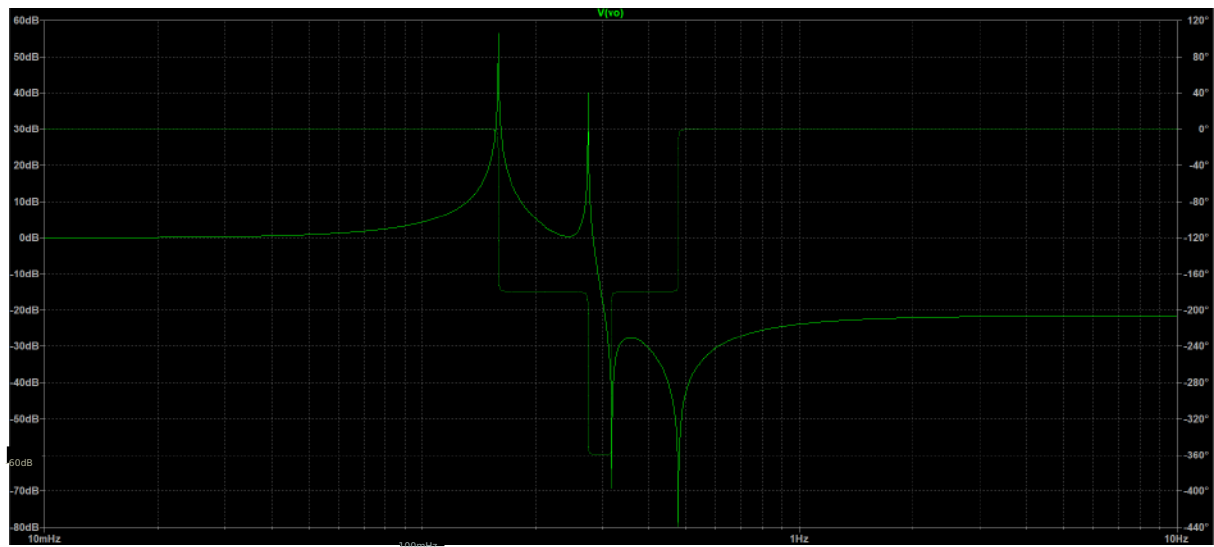


Chart 6

Output frequency response of the voltage transfer function.